

Concept 1 | Defining Angles and Basic Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Radians

$$\pi \text{ rad} = 180^\circ$$

Arc length

$$S = r\theta$$

Trig

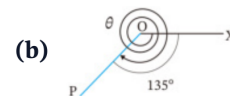
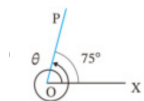
$$\sin \theta = \frac{y}{r}, \cos \theta = \frac{x}{r}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q001 Written

Find the measures of θ in the two figures: (a) one full counter-clockwise revolution plus 75° ; (b) two full clockwise revolutions and a further 135° .



Solution

1. Add $360^\circ + 75^\circ$ for (a).
2. Use $360^\circ \times (-2) - 135^\circ$ for (b).

Final answer

(a) 435° (b) -855°

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q002 Written

Draw the terminal side for 300° and -450° .

Solution

1. Reduce each angle by complete turns while keeping the sign of rotation.
2. 300° ends in quadrant IV; -450° is coterminal with -90° .

Final answer

300° : QIV -450° : negative y – axis

Q003 Written

Express 420° and -210° as $\alpha + 360^\circ \times n$, where n is an integer and $0^\circ \leq \alpha < 360^\circ$.

Solution

1. Choose the coterminal angle in the interval $0^\circ \leq \alpha < 360^\circ$.
2. Write the full family by adding $360^\circ \times n$.

Final answer

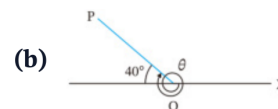
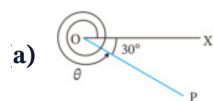
$60^\circ + 360^\circ \times n$; $150^\circ + 360^\circ \times n$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q004 Written

Find the measures of θ in the two figures: (a) a clockwise turn of 30° from the positive x -axis; (b) a terminal side 40° above the negative x -axis, measured counter-clockwise from the positive x -axis).



2 Draw the terminal side for the following angles.

Solution

1. Clockwise rotation is negative, so (a) is -30° .
2. For (b), $180^\circ - 40^\circ = 140^\circ$.

Final answer

(a) -30° (b) 140°

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q005 Written

Draw the terminal side for 500° and -120° .

Solution

1. Subtract or add complete turns: $500^\circ - 360^\circ = 140^\circ$; -120° is already in QIII.

Final answer

500° : *QI* -120° : *QIII*

Q006 Written

Express 520° and -140° as $\alpha + 360^\circ \times n$.

Solution

1. Reduce each angle to the principal representative in $[0^\circ, 360^\circ)$.
2. Attach $360^\circ n$.

Final answer

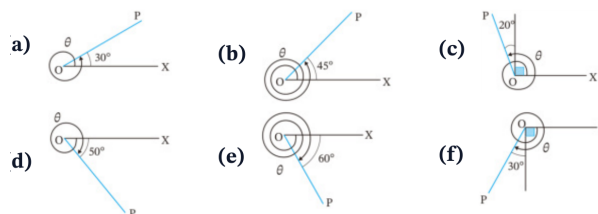
$160^\circ + 360^\circ \times n$; $220^\circ + 360^\circ \times n$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q007 Written

Find the six diagram measures θ shown in the given figure (a)–(f).



2 Draw the terminal side for the following angles.

Solution

1. Read the marked acute angle and the direction of each complete turn.
2. Use $360^\circ \pm$ marked angle and $90^\circ \pm 20^\circ$ where shown.

Final answer

30° ; 405° ; 110° ; 310° ; 300° ; 240°

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q008 Written

Draw the terminal sides for 470° , 780° , 225° , -380° , -990° , -200° .**Solution**

1. Reduce each angle modulo 360° , preserving clockwise direction for negative angles.
2. Sketch the ray from the positive x -axis in the resulting quadrant.

Final answer

QII; QII; QIII; QI; QII; QII

Q009 Written

Express 610° , 920° , 840° , -430° , -960° , -20° as $\alpha + 360^\circ \times n$.**Solution**

1. Find each representative α in $[0^\circ, 360^\circ)$.
2. The integer coefficient of 360° is absorbed into arbitrary integer n .

Final answer $250^\circ + 360^\circ n$; $200^\circ + 360^\circ n$; $120^\circ + 360^\circ n$; $290^\circ + 360^\circ n$; $120^\circ + 360^\circ n$; $340^\circ + 360^\circ n$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q010 **Written****Express 129720° and -2592100° as $\alpha + 360^\circ \times n$, with $0^\circ \leq \alpha < 360^\circ$.****Solution**

1. Divide by 360° and retain the remainder in the required interval.
2. Write each result as $\alpha + 360^\circ n$.

Final answer

$$120^\circ + 360^\circ n \quad ; \quad 260^\circ + 360^\circ n$$

Q011 **MCQ****The principal angle coterminal with 765° is:****Solution**

$$1. \quad 765^\circ - 720^\circ = 45^\circ$$

Correct option: A**Final answer**

$$45^\circ$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q012 MCQ

The principal angle coterminal with -410° is:**Solution**

$$1. -410^\circ + 720^\circ = 310^\circ$$

Correct option: C

Final answer 310°

Q013 MCQ

The terminal side of 540° lies on the:**Solution**

$$1. 540^\circ - 360^\circ = 180^\circ$$

Correct option: B

Final answernegative x -axis

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q014 MCQ

A positive angle of 315° terminates in:**Solution**1. $315^\circ \in \text{QIV}$

Correct option: D

Final answer

QIV

Q015 MCQ

Which is the general-angle form for -75° ?**Solution**1. $-75^\circ + 360^\circ = 285^\circ$

Correct option: B

Final answer $285^\circ + 360^\circ n$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q016 MCQ

The direction of a negative angle is:**Solution**

1. Negative rotation is clockwise.

Correct option: B

Final answer

clockwise

Q017 MCQ

The terminal side of 1080° is the:**Solution**1. $1080^\circ = 3 \times 360^\circ$

Correct option: A

Final answerpositive x -axis

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q018 MCQ

For $\alpha = 0^\circ$, the general-angle family is:**Solution**

1. Substitute $\alpha = 0^\circ$ in $\alpha + 360^\circ n$.

Correct option: A

Final answer $360^\circ n$

Q019 MCQ

The principal angle for -990° is:**Solution**

1. $-990^\circ + 1080^\circ = 90^\circ$

Correct option: C

Final answer 90°

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q020 MCQ

A ray (P) is 25° above the negative x -axis. Its standard-position angle is:

Solution

1. $180^\circ - 25^\circ = 155^\circ$

Correct option: B

Final answer

155°

Q021 Written

(1) Convert 630° to radians. (2) Convert $-\frac{7\pi}{4}$ to degrees. (3) For a sector with $r = 3$ and $\theta = \frac{4\pi}{3}$, find arc length L and area S .

Solution

1. Multiply degrees by $\pi/180$, and radians by $180/\pi$.
2. Use $L = r\theta$ and $S = \frac{1}{2}r^2\theta$ with θ in radians.

Final answer

$\frac{7\pi}{2}$; -315° ; $L = 4\pi$, $S = 6\pi$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q022 Written

Complete the standard-angle degree/radian table, then: convert $288^\circ, -400^\circ$ to radians; convert $\frac{11\pi}{6}, -\frac{3\pi}{5}$ to degrees; and find L, S for $r = 6, \theta = \frac{3\pi}{4}$.

Solution

1. Use the conversion multipliers and the standard table.
2. For a sector, substitute directly in $L = r\theta$ and $S = \frac{1}{2}r^2\theta$.

Final answer

$$\frac{8\pi}{5}, -\frac{20\pi}{9}; 330^\circ, -108^\circ; L = \frac{9\pi}{2}, S = \frac{27\pi}{2}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q023 Written

Convert $108^\circ, 36^\circ, -240^\circ, -225^\circ, 420^\circ, 400^\circ, -570^\circ, -2700^\circ$ to radians.

Solution

1. Multiply every degree measure by $\pi/180$.
2. Reduce each fraction to lowest terms.

Final answer

$$\frac{3\pi}{5}, \frac{\pi}{5}, -\frac{4\pi}{3}, -\frac{5\pi}{4}, \frac{7\pi}{3}, \frac{20\pi}{9}, -\frac{19\pi}{6}, -15\pi$$

Q024 Written

Convert $\frac{7\pi}{6}, \frac{4\pi}{5}, \frac{2\pi}{3}, \frac{4\pi}{3}, -\frac{5\pi}{4}, -\frac{\pi}{2}, -\frac{11\pi}{2}, -\frac{7\pi}{18}$ to degrees.

Solution

1. Multiply every radian measure by $180^\circ/\pi$.
2. Keep the sign and simplify.

Final answer

$$210^\circ, 144^\circ, 120^\circ, 240^\circ, -225^\circ, -90^\circ, -990^\circ, -70^\circ$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q025 Written

Find L and S for: (a) $r = 6, \theta = \frac{7\pi}{6}$; (b) $r = 4, \theta = \frac{2\pi}{5}$; (c) $r = 6, \theta = \frac{5\pi}{4}$.

Solution

1. Apply $L = r\theta$.
2. Apply $S = \frac{1}{2}r^2\theta$ to each sector.

Final answer

(a) $(7\pi, 21\pi)$; (b) $(\frac{8\pi}{5}, \frac{16\pi}{5})$; (c) $(\frac{15\pi}{2}, \frac{45\pi}{2})$

Q026 Written

Draw terminal sides for $\frac{\pi}{4}, \frac{\pi}{3}, \frac{5\pi}{6}, \frac{3\pi}{4}, \frac{4\pi}{3}, \frac{7\pi}{6}, -\frac{\pi}{3}, -\frac{\pi}{4}$.

Solution

1. Use the standard radian-angle table and the sign of rotation.
2. Sketch each terminal ray from the positive x -axis.

Final answer

QI; QI; QII; QII; QIII; QIII; QIV; QIV

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q027 Written

A sector of radius 9 cm and central angle $4\pi/3$ is joined to form a cone. Determine the exact vertical height and total volume.

Solution

1. The sector arc becomes the cone base circumference: $2\pi R = 9(4\pi/3)$, so $R = 6$.
2. The slant height is 9, hence $h = \sqrt{9^2 - 6^2} = 3\sqrt{5}$.
3. Use $V = \frac{1}{3}\pi R^2 h$.

Final answer

$$h = 3\sqrt{5} \text{ cm}; \quad V = 36\pi\sqrt{5} \text{ cm}^3$$

Q028 MCQ

Convert 150° to radians:

Solution

1. $150 \times \pi/180 = 5\pi/6$

Correct option: A

Final answer

$$\frac{5\pi}{6}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q029 MCQ

Convert $-\frac{5\pi}{6}$ to degrees:

Solution

1. $-5\pi/6 \times 180/\pi = -150^\circ$

Correct option: B

Final answer

-150°

Q030 MCQ

For $r = 5, \theta = \frac{2\pi}{3}$, the arc length is:

Solution

1. $L = r\theta = 5(2\pi/3)$

Correct option: B

Final answer

$\frac{10\pi}{3}$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q031 MCQ

For $r = 4$, $\theta = \frac{\pi}{2}$, the sector area is:

Solution

1. $S = \frac{1}{2}(4^2)(\pi/2) = 4\pi$

Correct option: B

Final answer

4π

Q032 MCQ

The radian measure of a full turn is:

Solution

1. A full circumference subtends 2π radians.

Correct option: B

Final answer

2π

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q033 MCQ

The degree measure of $\frac{7\pi}{4}$ is:

Solution

1. $7\pi/4 \times 180/\pi = 315^\circ$

Correct option: C

Final answer

315°

Q034 MCQ

The principal angle for $-\frac{11\pi}{6}$ is:

Solution

1. $-11\pi/6 + 2\pi = \pi/6$

Correct option: A

Final answer

$\frac{\pi}{6}$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q035 MCQ

For $r = 3$, $\theta = \pi$, the sector area is:**Solution**

1. $S = \frac{1}{2}(3^2)\pi = 9\pi/2$

Correct option: D**Final answer**

$$\frac{9\pi}{2}$$

Q036 MCQ

The angle $\frac{5\pi}{3}$ lies in:**Solution**

1. $5\pi/3 = 300^\circ$

Correct option: D**Final answer**

QIV

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q037 Written

Convert 252° to radians.**Solution**

$$1. 252\pi/180 = 7\pi/5.$$

Final answer

$$\frac{7\pi}{5}$$

Q038 Written

A sector has $r = 8$ and $\theta = \frac{3\pi}{4}$. Find L and S .**Solution**

$$1. L = 8(3\pi/4) = 6\pi.$$

$$2. S = \frac{1}{2}(8^2)(3\pi/4) = 24\pi.$$

Final answer

$$L = 6\pi, \quad S = 24\pi$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q039 Written

If $\theta = \frac{4\pi}{3}$, find $\sin \theta$, $\cos \theta$, $\tan \theta$.

Solution

1. The terminal side is in QIII with reference angle $\pi/3$.
2. Use the signs of x, y on the unit circle.

Final answer

$$-\frac{\sqrt{3}}{2}, -\frac{1}{2}, \sqrt{3}$$

Q040 Written

Find the exact values (write undefined where appropriate): $\sin \frac{7\pi}{4}$, $\cos \frac{\pi}{3}$, $\tan \frac{\pi}{6}$, $\sin(-\frac{2\pi}{3})$, $\cos(-\pi)$, $\tan(-\frac{\pi}{2})$.

Solution

1. Locate each angle on the unit circle.
2. Read x, y , then use $\sin \theta = y$, $\cos \theta = x$, $\tan \theta = y/x$.

Final answer

$$-\frac{\sqrt{2}}{2}; \frac{1}{2}; \frac{1}{\sqrt{3}}; -\frac{\sqrt{3}}{2}; -1; \text{undefined}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q041 Written

Find the exact values of the eighteen trigonometric functions listed in the given exercise (a)–(r), including negative and undefined cases.

Find the values of the following trigonometric functions. If the value is undefined, write “undefined”.

(a) $\sin \frac{\pi}{3}$

(b) $\sin \frac{7}{6}\pi$

(c) $\sin \pi$

(d) $\cos \frac{5}{4}\pi$

(e) $\cos \frac{5}{3}\pi$

(f) $\cos \frac{7}{6}\pi$

(g) $\tan \frac{2}{3}\pi$

(h) $\tan \frac{5}{4}\pi$

(i) $\tan \pi$

(j) $\sin \left(-\frac{\pi}{6}\right)$

(k) $\sin \left(-\frac{5}{4}\pi\right)$

(l) $\sin \left(-\frac{5}{3}\pi\right)$

(m) $\cos \left(-\frac{\pi}{4}\right)$

(n) $\cos \left(-\frac{5}{6}\pi\right)$

(o) $\cos \left(-\frac{4}{3}\pi\right)$

(p) $\tan \left(-\frac{4}{3}\pi\right)$

(q) $\tan \left(-\frac{11}{6}\pi\right)$

(r) $\tan \left(-\frac{3}{2}\pi\right)$

Solution

1. Reduce each angle to a standard reference angle.
2. Apply quadrant signs and the unit-circle exact values; tangent is undefined when $x = 0$.

Final answer

$\frac{\sqrt{3}}{2}; -\frac{1}{2}; 0; -\frac{\sqrt{2}}{2}; \frac{1}{2}; -\frac{\sqrt{3}}{2}; -\sqrt{3}; 1; 0; -\frac{1}{2}; \frac{\sqrt{2}}{2}; \frac{\sqrt{3}}{2}; \frac{\sqrt{2}}{2}; -\frac{\sqrt{3}}{2}; -\frac{1}{2}; -\sqrt{3}; \frac{1}{\sqrt{3}}; \text{undefined}$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q042 **Written**

The panel height is $h(t) = 3 \sin(\frac{2\pi}{3}t - \frac{\pi}{4}) + 1.5$. Find the exact height at $t = 0$.

Solution

1. Substitute $t = 0$: $h(0) = 3 \sin(-\pi/4) + 1.5$.
2. Use $\sin(-\pi/4) = -\sqrt{2}/2$.

Final answer

$$\frac{3(1 - \sqrt{2})}{2} \text{ m}$$

Q043 **MCQ**

$\sin \frac{\pi}{6}$ equals:

Solution

1. Use the QI special-angle value.

Correct option: B

Final answer

$$\frac{1}{2}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q044 MCQ

 $\cos \frac{2\pi}{3}$ equals:**Solution**1. QII has negative cosine; reference angle $\pi/3$.

Correct option: B

Final answer

$$-\frac{1}{2}$$

Q045 MCQ

 $\tan \frac{3\pi}{4}$ equals:**Solution**1. QII reference angle $\pi/4$.

Correct option: A

Final answer

$$-1$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q046 MCQ

 $\sin(-\frac{\pi}{2})$ equals:**Solution**1. The terminal point is $(0, -1)$.

Correct option: A

Final answer

-1

Q047 MCQ

 $\tan \frac{\pi}{2}$ is:**Solution**1. $\tan = y/x$ and $x = 0$.

Correct option: D

Final answer

undefined

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q048 MCQ

 $\cos \frac{5\pi}{3}$ equals:**Solution**1. $5\pi/3$ has reference angle $\pi/3$ in QIV.

Correct option: D

Final answer

$$\frac{1}{2}$$

Q049 MCQ

If $P(x, y)$ lies on the unit circle, then $\sin \theta$ is:**Solution**1. By definition, $\sin \theta = y$.

Correct option: B

Final answer

$$y$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q050 MCQ

 $\tan\left(-\frac{3\pi}{4}\right)$ equals:**Solution**1. Tangent is odd: $\tan(-3\pi/4) = -\tan(3\pi/4) = 1$.

Correct option: C

Final answer

1

Q051 MCQ

 $\sin \frac{11\pi}{6}$ equals:**Solution**1. QIV reference angle $\pi/6$, so sine is negative.

Correct option: A

Final answer $-\frac{1}{2}$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q052 Written

Find $\sin \frac{5\pi}{6}, \cos \frac{4\pi}{3}, \tan \frac{7\pi}{4}$.**Solution**

1. Use reference angles $\pi/6, \pi/3, \pi/4$ and quadrant signs.

Final answer

$$\frac{1}{2}, -\frac{1}{2}, -1$$

Q053 Written

Find $\sin(-\frac{5\pi}{6}), \cos(-\frac{7\pi}{6}), \tan(-\frac{\pi}{3})$.**Solution**

1. Reduce negative angles and apply odd/even symmetry.

Final answer

$$-\frac{1}{2}, -\frac{\sqrt{3}}{2}, -\sqrt{3}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q054 Written

For $P = (-\frac{1}{2}, \frac{\sqrt{3}}{2})$ on the unit circle, find the corresponding $\sin \theta$, $\cos \theta$, $\tan \theta$.

Solution

1. Read x, y directly, then compute y/x .

Final answer

$$\frac{\sqrt{3}}{2}, -\frac{1}{2}, -\sqrt{3}$$

Q055 Written

Find the exact value of $3 \sin \frac{3\pi}{2} - 2 \cos \pi$.

Solution

1. $3(-1) - 2(-1) = -1$.

Final answer

$$-1$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q056 Written

State the two standard angles where $\tan \theta$ is undefined in $[0, 2\pi)$.**Solution**

1. Tangent is undefined when the unit-circle x -coordinate is zero.

Final answer

$$\frac{\pi}{2}, \frac{3\pi}{2}$$

Q057 Written

Solve for θ , $0 \leq \theta < 2\pi$: (a) $\sin \theta = -\frac{1}{2}$; (b) $\cos \theta = \frac{1}{\sqrt{2}}$; (c) $\tan \theta = -\sqrt{3}$.**Solution**

1. Draw the unit circle and the corresponding horizontal/vertical/oblique line.
2. Use the reference angle and select the quadrants with the required sign.

Final answer

$$(a) \frac{7\pi}{6}, \frac{11\pi}{6}; (b) \frac{\pi}{4}, \frac{7\pi}{4}; (c) \frac{2\pi}{3}, \frac{5\pi}{3}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q058 Written

Solve the six trigonometric equations in the given Try set for $0 \leq \theta < 2\pi$: $\sin \theta = 1/\sqrt{2}$, $\cos \theta = -\sqrt{3}/2$, $\tan \theta = 1/\sqrt{3}$, $\sin \theta = 0$, $\cos \theta = 1/2$, $\tan \theta = -1$.

Solution

1. Find the reference angle from the special triangle.
2. Use the sign of the function to choose the correct quadrants.

Final answer

$$\frac{\pi}{4}, \frac{3\pi}{4}; \frac{5\pi}{6}, \frac{7\pi}{6}; \frac{\pi}{6}, \frac{7\pi}{6}; 0, \pi; \frac{\pi}{3}, \frac{5\pi}{3}; \frac{3\pi}{4}, \frac{7\pi}{4}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q059 Written

Solve the sixteen equations in the given exercise set (a)–(p) for $0 \leq \theta < 2\pi$.

Solution

1. Apply the same unit-circle/reference-angle method to each equation.
2. Remember that sine/cosine can have two solutions, while $\sin \theta = \pm 1$ and $\cos \theta = \pm 1$ have one in the interval.

Final answer

$$\frac{\pi}{6}, \frac{5\pi}{6}; \frac{\pi}{3}, \frac{2\pi}{3}; \frac{\pi}{2}, \frac{3\pi}{2}; 0; 0, \pi; \frac{\pi}{3}, \frac{4\pi}{3}; \frac{\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}; \frac{3\pi}{4}, \frac{5\pi}{4}; \frac{2\pi}{3}, \frac{4\pi}{3}; \frac{5\pi}{6}, \frac{11\pi}{6}; \frac{\pi}{4}, \frac{5\pi}{4}; \frac{4\pi}{3}, \frac{5\pi}{3}; \frac{3\pi}{2}; \pi; \frac{\pi}{6}, \frac{11\pi}{6}$$

Q060 Written

Solve $\tan^2 \theta + (1 - \sqrt{3}) \tan \theta = \sqrt{3}$ for $0 \leq \theta < 360^\circ$.

Solution

1. Let $u = \tan \theta$ and factor $u^2 + (1 - \sqrt{3})u - \sqrt{3} = 0$.
2. The two tangent values are 1 and $-\sqrt{3}$; list their angles in $[0, 360^\circ)$.

Final answer

$$60^\circ, 135^\circ, 240^\circ, 315^\circ$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q061 MCQ

For $0 \leq \theta < 2\pi$, $\sin \theta = \frac{1}{2}$ gives:

Solution

1. Sine is positive in QI and QII with reference angle $\pi/6$.

Correct option: A

Final answer

$$\frac{\pi}{6}, \frac{5\pi}{6}$$

Q062 MCQ

For $0 \leq \theta < 2\pi$, $\cos \theta = -\frac{1}{2}$ gives:

Solution

1. Cosine is negative in QII and QIII.

Correct option: B

Final answer

$$\frac{2\pi}{3}, \frac{4\pi}{3}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q063 MCQ

For $0 \leq \theta < 2\pi$, $\tan \theta = 1$ gives:**Solution**

1. Tangent is positive in QI and QIII.

Correct option: A**Final answer**

$$\frac{\pi}{4}, \frac{5\pi}{4}$$

Q064 MCQ

The solutions of $\sin \theta = 0$ in $[0, 2\pi)$ are:**Solution**

1. The unit-circle y-coordinate is zero on the x-axis.

Correct option: A**Final answer**

$$0, \pi$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q065 MCQ

If $\cos \theta = 1$, then in $[0, 2\pi)$, θ is:

Solution

1. The point $(1, 0)$ corresponds to $\theta = 0$.

Correct option: A

Final answer

0

Q066 MCQ

For $0 \leq \theta < 2\pi$, $\tan \theta = -\sqrt{3}$ gives:

Solution

1. Tangent is negative in QII/QIV; reference angle $\pi/3$.

Correct option: B

Final answer

$\frac{2\pi}{3}, \frac{5\pi}{3}$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q067 MCQ

For $0 \leq \theta < 2\pi$, $\sin \theta = -1$ gives:**Solution**

1. The lowest unit-circle point has y-coordinate -1 .

Correct option: B**Final answer**

$$\frac{3\pi}{2}$$

Q068 MCQ

For $0 \leq \theta < 2\pi$, $\cos \theta = 0$ gives:**Solution**

1. Cosine is the x-coordinate, zero on the y-axis.

Correct option: B**Final answer**

$$\frac{\pi}{2}, \frac{3\pi}{2}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q069 MCQ

The equation $\tan \theta = 0$ has solutions in $[0, 2\pi)$:

Solution

1. Tangent is zero when $y = 0$ and $x \neq 0$.

Correct option: A

Final answer

$0, \pi$

Q070 Written

Solve $\cos \theta = \frac{\sqrt{3}}{2}$ for $0 \leq \theta < 2\pi$.

Solution

1. Cosine is positive in QI/QIV with reference angle $\pi/6$.

Final answer

$\frac{\pi}{6}, \frac{11\pi}{6}$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q071 Written

Solve $\sin \theta = -\frac{\sqrt{3}}{2}$ for $0 \leq \theta < 2\pi$.

Solution

1. Sine is negative in QIII/QIV with reference angle $\pi/3$.

Final answer

$$\frac{4\pi}{3}, \frac{5\pi}{3}$$

Q072 Written

Solve $\tan \theta = \frac{1}{\sqrt{3}}$ for $0 \leq \theta < 2\pi$.

Solution

1. Tangent is positive in QI/QIII with reference angle $\pi/6$.

Final answer

$$\frac{\pi}{6}, \frac{7\pi}{6}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Q073 Written

Solve $2 \cos \theta - 1 = 0$ for $0 \leq \theta < 2\pi$.**Solution**1. Rearrange to $\cos \theta = 1/2$, then choose QI/QIV.**Final answer**

$$\frac{\pi}{3}, \frac{5\pi}{3}$$

Q074 Written

Solve $2 \sin \theta + \sqrt{2} = 0$ for $0 \leq \theta < 2\pi$.**Solution**1. Rearrange to $\sin \theta = -\sqrt{2}/2$, then choose QIII/QIV.**Final answer**

$$\frac{5\pi}{4}, \frac{7\pi}{4}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Quiz MCQ 1 **Concept Quiz · MCQ**

The radian measure of 240° is:

Solution

1. $240\pi/180 = 4\pi/3$.

Correct option: B

Final answer

$$\frac{4\pi}{3}$$

Quiz MCQ 2 **Concept Quiz · MCQ**

If $\sin \theta = -\frac{\sqrt{3}}{2}$, $0 \leq \theta < 2\pi$, then θ is:

Solution

1. Sine is negative in QIII/QIV with reference angle $\pi/3$.

Correct option: B

Final answer

$$\frac{4\pi}{3}, \frac{5\pi}{3}$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Quiz MCQ 3 **Concept Quiz · MCQ**

For a sector with $r = 6$ and $\theta = \frac{\pi}{3}$, the area is:

Solution

1. $S = \frac{1}{2}(6^2)(\pi/3) = 6\pi$.

Correct option: C

Final answer

$$6\pi$$

Quiz MCQ 4 **Concept Quiz · MCQ**

The general-angle family for -300° is:

Solution

1. Add 360° to obtain the principal angle 60° .

Correct option: A

Final answer

$$60^\circ + 360^\circ n$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Quiz WRITTEN 5 Concept Quiz · Written

Solve $\cos \theta = -\frac{\sqrt{2}}{2}$ for $0 \leq \theta < 2\pi$.

Solution

1. Cosine is negative in QII/QIII with reference angle $\pi/4$.

Final answer

$$\frac{3\pi}{4}, \frac{5\pi}{4}$$

Quiz WRITTEN 6 Concept Quiz · Written

Convert $-\frac{13\pi}{6}$ to degrees and state its principal angle.

Solution

1. Multiply by $180/\pi$: -390° . Add 720° for the principal representative 330° .

Final answer

$$-390^\circ; 330^\circ$$

Worked Solutions

Defining Angles and Basic Trigonometric Functions

Quiz WRITTEN 7 Concept Quiz · Written

Find L and S for a sector with $r = 10$ and $\theta = \frac{2\pi}{5}$.

Solution

1. $L = 10(2\pi/5) = 4\pi$.
2. $S = \frac{1}{2}(10^2)(2\pi/5) = 20\pi$.

Final answer

$$L = 4\pi, \quad S = 20\pi$$

Quiz WRITTEN 8 Concept Quiz · Written

Evaluate $3 \sin \frac{7\pi}{6} - 2 \cos \frac{4\pi}{3}$.

Solution

1. $3(-1/2) - 2(-1/2) = -1/2$.

Final answer

$$-\frac{1}{2}$$

Answer key

Use the question number shown on each page.

Q001 (a) 435° (b) -855°

Q002 300° : QIV

-450° : *negative y - axis*

Q003 $60^\circ + 360^\circ \times n$; $150^\circ + 360^\circ \times n$

Q004 (a) -30° (b) 140°

Q005 500° : QI -120° : QIII

Q006 $160^\circ + 360^\circ \times n$; $220^\circ + 360^\circ \times n$

Q007 30° ; 405° ; 110° ; 310° ; 300° ; 240°

Q008 QII; QII; QIII; QI; QII; QII

Q009 $250^\circ + 360^\circ n$; $200^\circ + 360^\circ n$; $120^\circ + 360^\circ n$; $290^\circ + 360^\circ n$; $120^\circ + 360^\circ n$; $340^\circ + 360^\circ n$

Q010 $120^\circ + 360^\circ n$; $260^\circ + 360^\circ n$

Q011 A

Q012 C

Q013 B

Q014 D

Q015 B

Q016 B

Q017 A

Q018 A

Q019 C

Q020 B

Q021 $\frac{7\pi}{2}$; -315° ; $L = 4\pi$, $S = 6\pi$

Q022 $\frac{8\pi}{5}$, $-\frac{20\pi}{9}$; 330° , -108° ; $L = \frac{9\pi}{2}$, $S = \frac{27\pi}{2}$

Q023 $\frac{3\pi}{5}$, $\frac{\pi}{5}$, $-\frac{4\pi}{3}$, $-\frac{5\pi}{4}$, $\frac{7\pi}{3}$, $\frac{20\pi}{9}$, $-\frac{19\pi}{6}$, -15π

Q024 210° , 144° , 120° , 240° , -225° , -90° , -990° , -70°

Q025 (a) $(7\pi, 21\pi)$; (b) $(\frac{8\pi}{5}, \frac{16\pi}{5})$; (c) $(\frac{15\pi}{2}, \frac{45\pi}{2})$

Q026 QI; QI; QII; QII; QIII; QIII; QIV; QIV

Q027 $h = 3\sqrt{5}$ cm; $V = 36\pi\sqrt{5}$ cm³

Q028 A

Q029 B

Q030 B

Q031 B

Q032 B

Q033 C

Q034 A

Q035 D

Q036 D

Q037 $\frac{7\pi}{5}$

Q038 $L = 6\pi$, $S = 24\pi$

Q039 $-\frac{\sqrt{3}}{2}$, $-\frac{1}{2}$, $\sqrt{3}$

Q040 $-\frac{\sqrt{2}}{2}$; $\frac{1}{2}$; $\frac{1}{\sqrt{3}}$; $-\frac{\sqrt{3}}{2}$; -1 ; *undefined*

Q041 $\frac{\sqrt{3}}{2}$; $-\frac{1}{2}$; 0 ; $-\frac{\sqrt{2}}{2}$; $\frac{1}{2}$; $-\frac{\sqrt{3}}{2}$; $-\sqrt{3}$; 1 ; 0 ; $-\frac{1}{2}$; $\frac{\sqrt{2}}{2}$; $\frac{\sqrt{3}}{2}$; $-\frac{\sqrt{3}}{2}$; $-\frac{1}{2}$; $-\sqrt{3}$; $\frac{1}{\sqrt{3}}$; *undefined*

Q042 $\frac{3(1-\sqrt{2})}{2}$ m

Q043 B

Q044 B

Q045 A

Q046 A

Q047 D

Q048 D

Q049 B

Q050 C

Q051 A

Q052 $\frac{1}{2}$, $-\frac{1}{2}$, -1

Q053 $-\frac{1}{2}$, $-\frac{\sqrt{3}}{2}$, $-\sqrt{3}$

Q054 $\frac{\sqrt{3}}{2}$, $-\frac{1}{2}$, $-\sqrt{3}$

Q055 -1

Q056 $\frac{\pi}{2}$, $\frac{3\pi}{2}$

Q057 (a) $\frac{7\pi}{6}$, $\frac{11\pi}{6}$; (b) $\frac{\pi}{4}$, $\frac{7\pi}{4}$; (c) $\frac{2\pi}{3}$, $\frac{5\pi}{3}$

Q058 $\frac{\pi}{4}$, $\frac{3\pi}{4}$; $\frac{5\pi}{6}$, $\frac{7\pi}{6}$; $\frac{\pi}{6}$, $\frac{7\pi}{6}$; 0 , π ; $\frac{\pi}{3}$, $\frac{5\pi}{3}$; $\frac{3\pi}{4}$, $\frac{7\pi}{4}$

Q059 $\frac{\pi}{6}$, $\frac{5\pi}{6}$; $\frac{\pi}{3}$, $\frac{2\pi}{3}$; $\frac{3\pi}{2}$, 0 ; 0 , π ; $\frac{\pi}{3}$, $\frac{4\pi}{3}$; $\frac{\pi}{2}$, $\frac{5\pi}{2}$; $\frac{\pi}{4}$, $\frac{5\pi}{4}$; $\frac{3\pi}{4}$, $\frac{7\pi}{4}$; $\frac{2\pi}{3}$, $\frac{4\pi}{3}$; $\frac{5\pi}{6}$, $\frac{11\pi}{6}$; $\frac{\pi}{4}$, $\frac{5\pi}{4}$; $\frac{3\pi}{2}$, $\frac{3\pi}{2}$; $\frac{\pi}{6}$, $\frac{11\pi}{6}$

Q060 60° , 135° , 240° , 315°

Q061 $\frac{1}{2}$, $\frac{\sqrt{2}}{2}$; $\frac{\sqrt{3}}{2}$; $\frac{\sqrt{2}}{2}$; $-\frac{\sqrt{3}}{2}$; $-\frac{1}{2}$; $-\sqrt{3}$; $\frac{1}{\sqrt{3}}$; *undefined*

Q062 B

Q063 A

Q064 A

Q065 A

Q066 B

Q067 B

Q068 B

Q069 A

Q070 $\frac{\pi}{6}$, $\frac{11\pi}{6}$

Q071 $\frac{4\pi}{3}$, $\frac{5\pi}{3}$

Q072 $\frac{\pi}{6}$, $\frac{7\pi}{6}$

Q073 $\frac{\pi}{3}$, $\frac{5\pi}{3}$

Q074 $\frac{5\pi}{4}$, $\frac{7\pi}{4}$

Quiz MCQ 1 B

Quiz MCQ 2 B

Quiz MCQ 3 C

Quiz MCQ 4 A

Quiz WRITTEN 5 $\frac{3\pi}{4}$, $\frac{5\pi}{4}$

Quiz WRITTEN 6 -390° ; 330°

Quiz WRITTEN 7 $L = 4\pi$, $S = 20\pi$

Quiz WRITTEN 8 $-\frac{1}{2}$